Dijkstra with a negative edge weight

Sensitivity: Internal

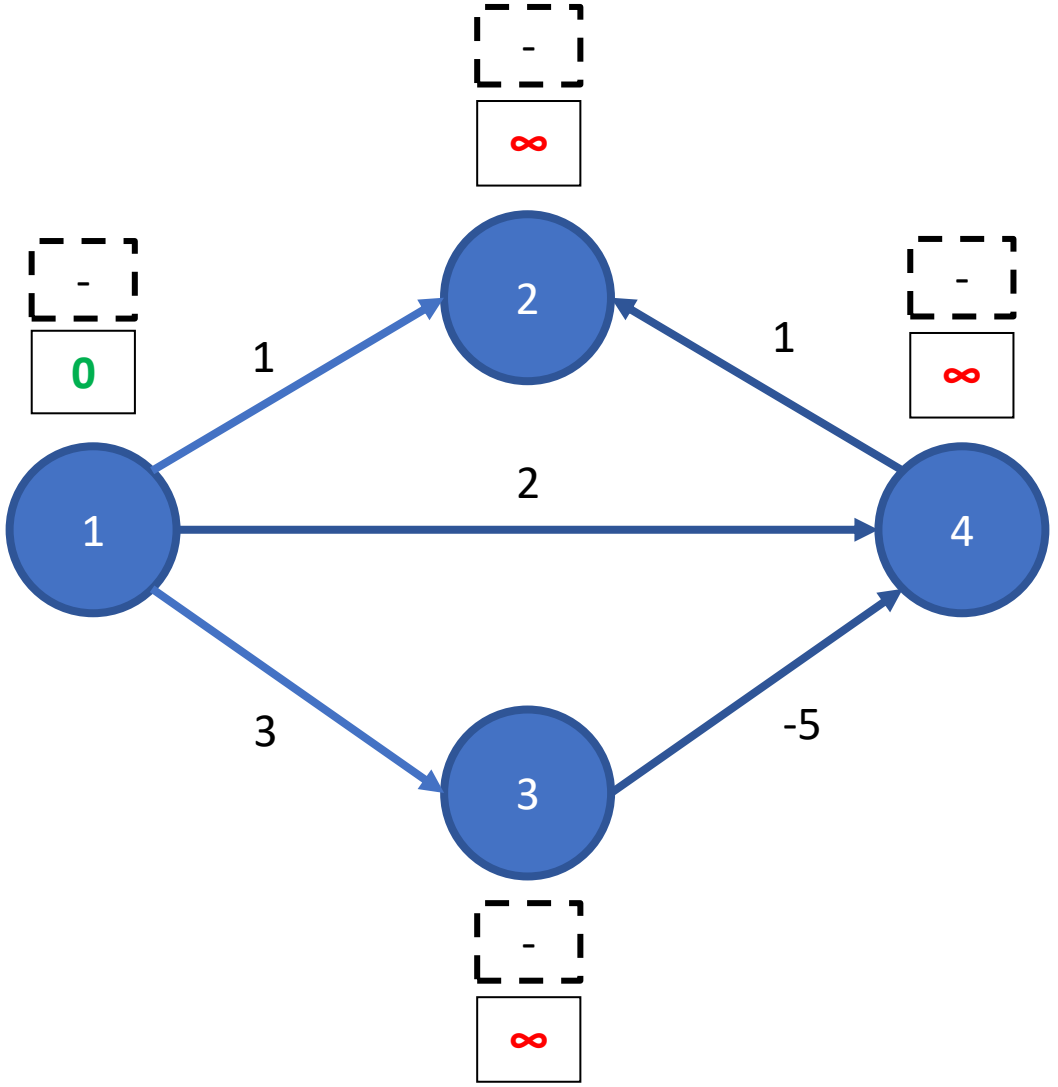


Permanent



Temporary

v	Label	d_i	π_i
1	T	0	-
2	T		
3	T		
4	T		



Sensitivity: Internal

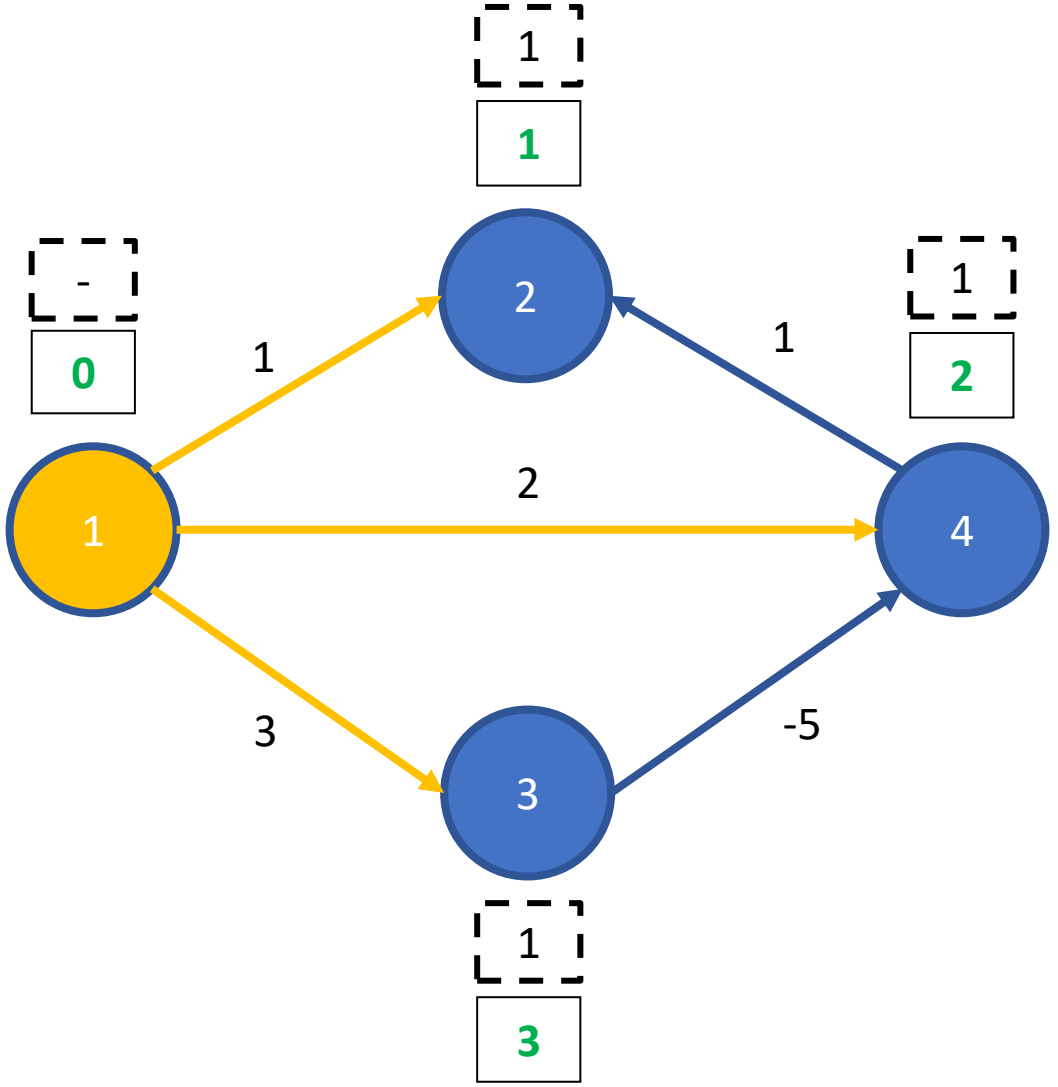


Permanent



Temporary

v	Label	d_i	π_i
1	P	0	-
2	T	1	1
3	T	3	1
4	T	2	1



Sensitivity: Internal

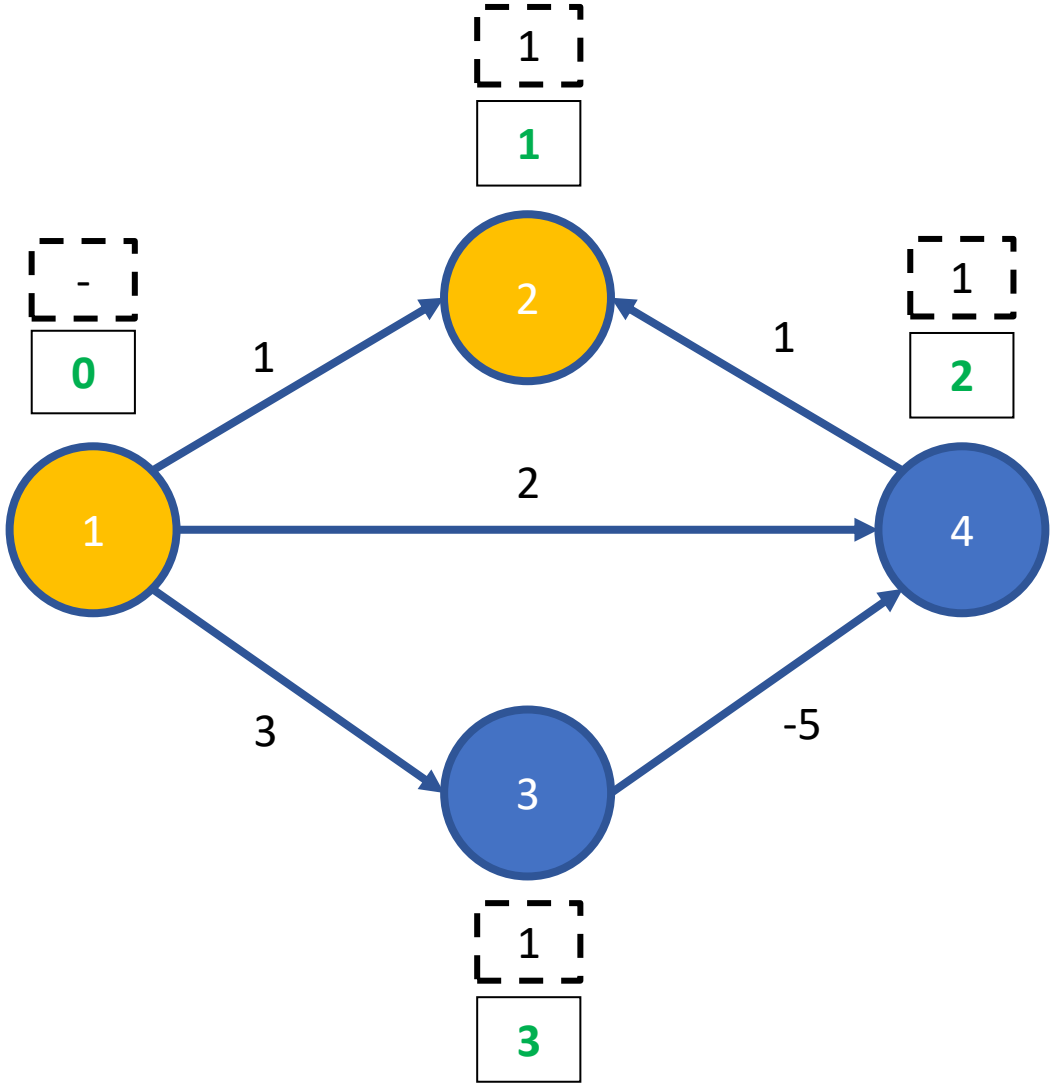


Permanent



Temporary

v	Label	d_i	π_i
1	P	0	-
2	P	1	1
3	T	3	1
4	T	2	1



Sensitivity: Internal

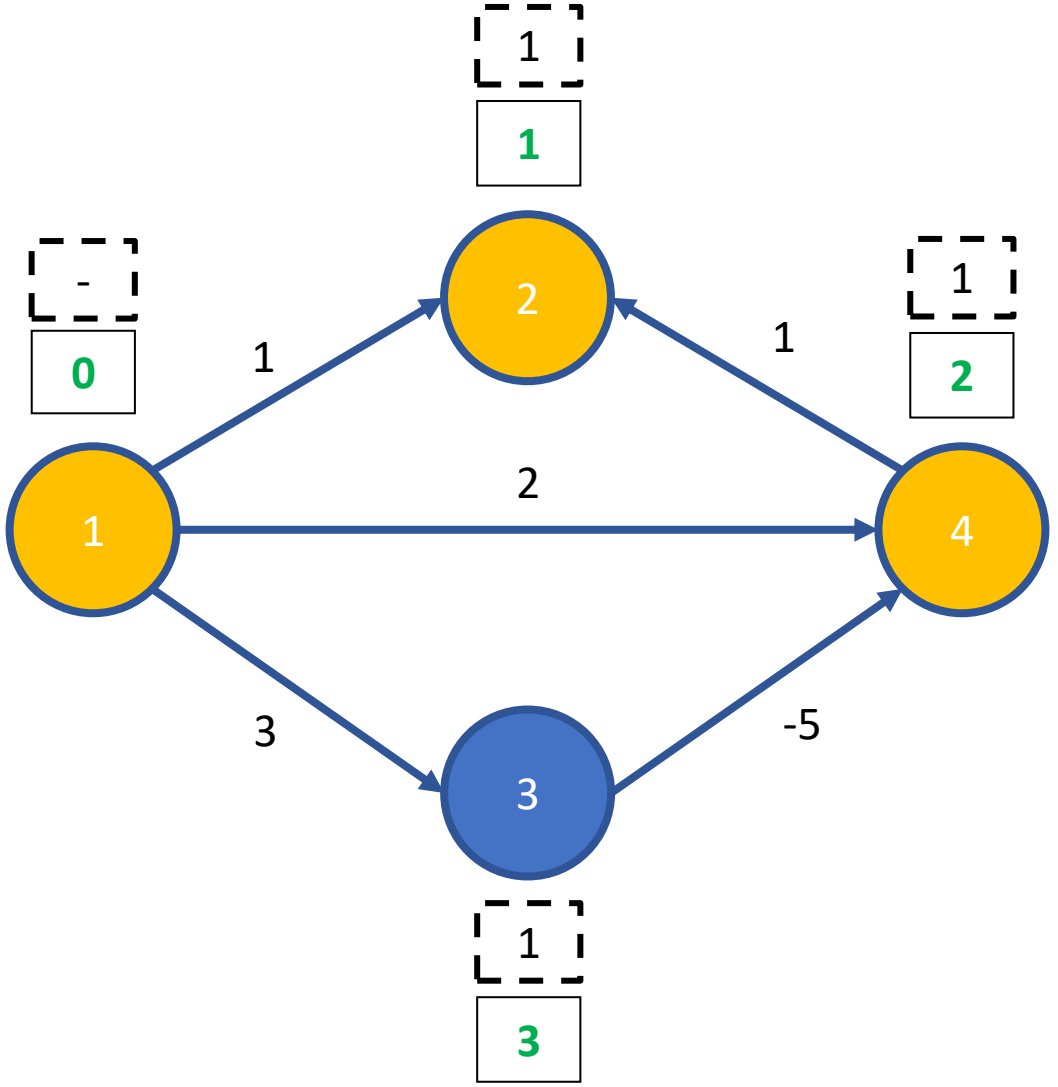


Permanent



Temporary

v	Label	d_i	π_i
1	P	0	-
2	P	1	1
3	T	3	1
4	P	2	1



Sensitivity: Internal

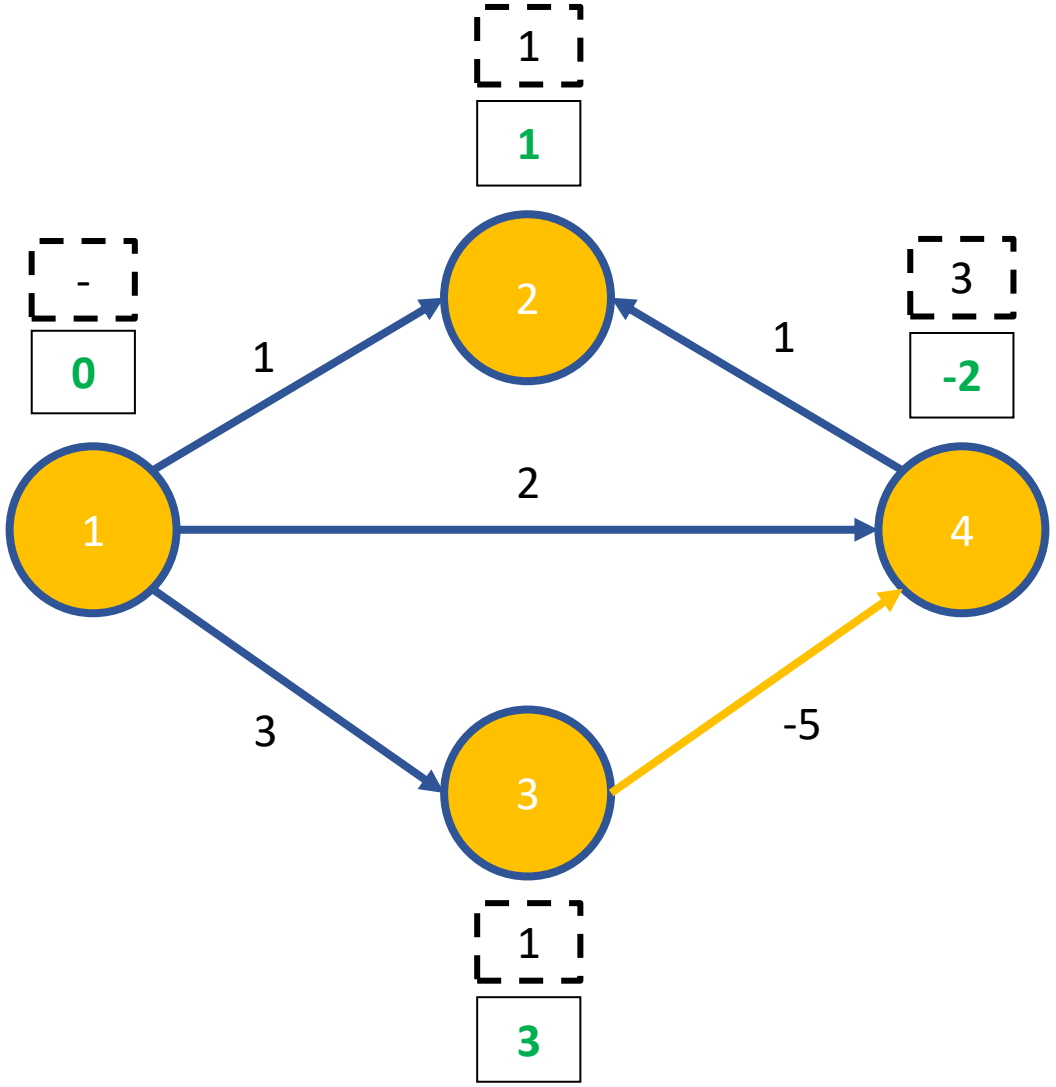


Permanent



Temporary

v	Label	d_i	π_i
1	P	0	-
2	P	1	1
3	P	3	1
4	P	-2	3



What is the best path from our table for 1 -> 2?

1-2 with distance 1

Is there a better path?

Yes 1-3-4-2 has a distance of -1.

Dijkstra delivered a suboptimal solution. i.e., it failed

Sensitivity: Internal